WEBER STATE UNIVERSITY
Engineering, Applied Science & Technology

DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

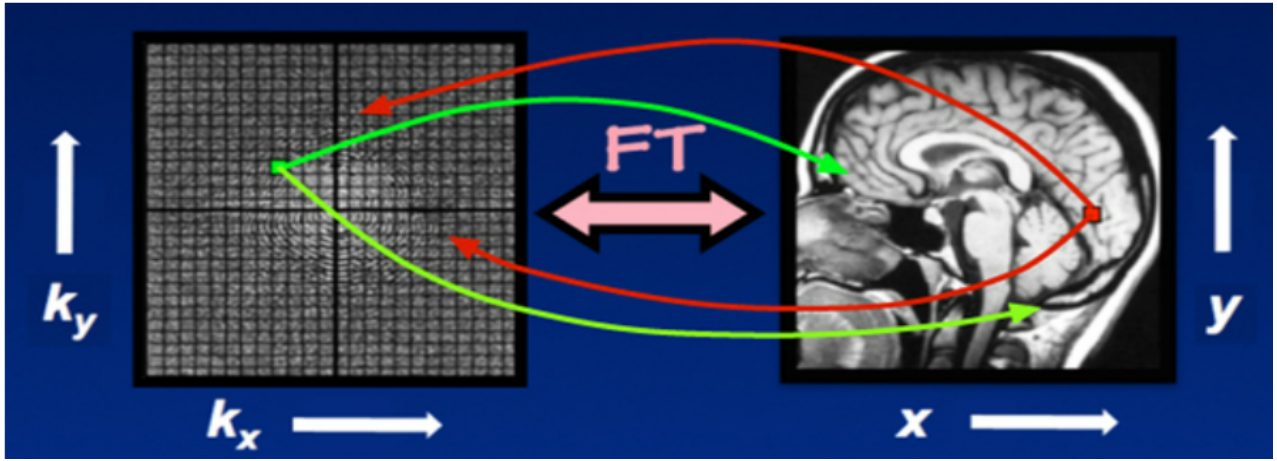
Reconstruction of Undersampled MRI Data to Reduce Scan Time

Feasibility Study and Business Case

Shreeya Maharjan ECE 3090 | Weber State University | February 25, 2026

How MRI works





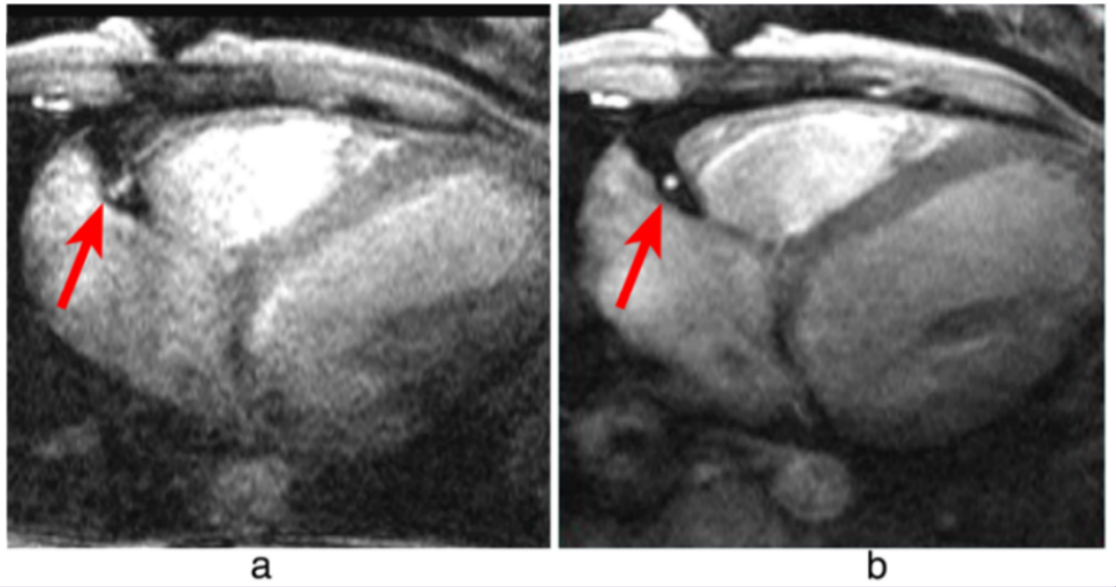
The Problem

MRI acquisition is slow

Scans can take several minutes, causing patient discomfort and motion artifacts.

Cardiac MRI is especially difficult

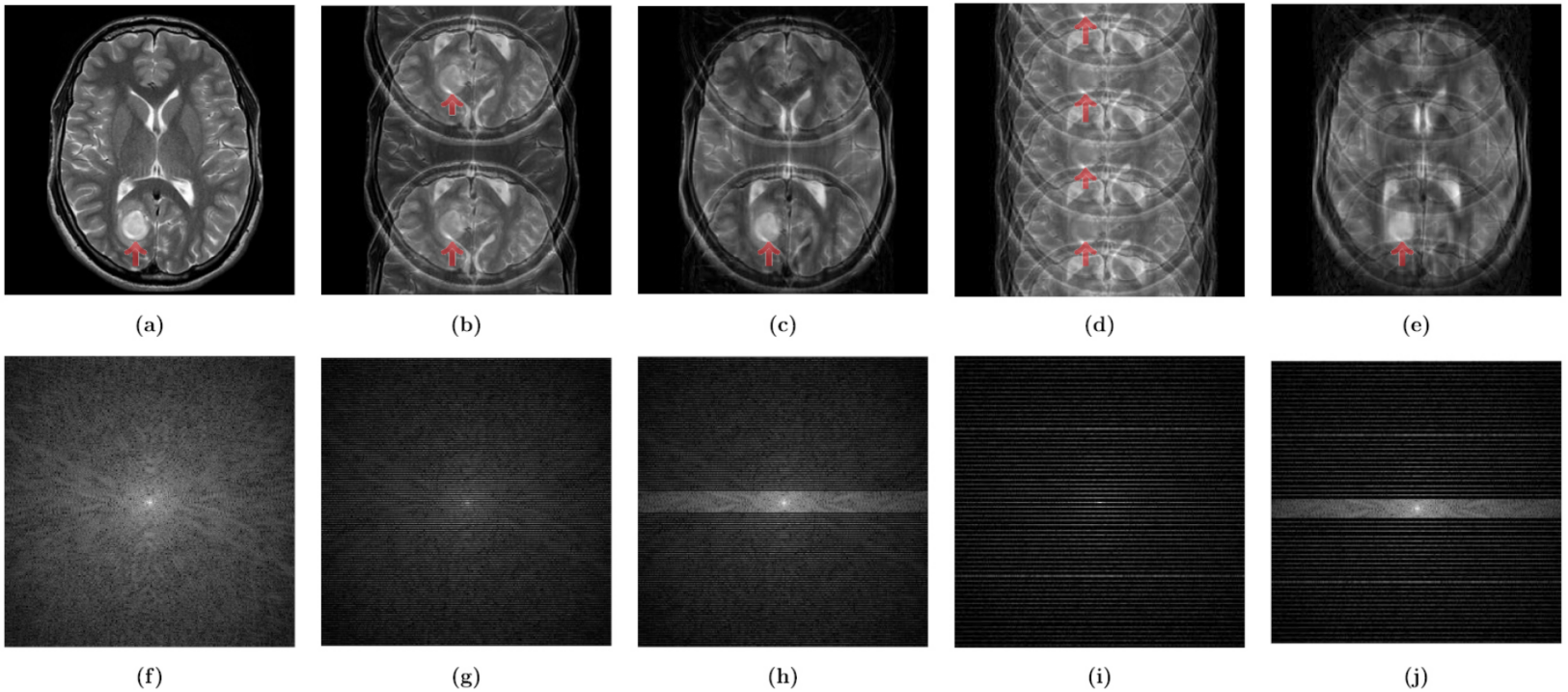
The heart's continuous motion causes blurring and ghosting artifacts



Undersampling offers a path forward

Collecting fewer k-space samples speeds up acquisition, but introduces aliasing artifacts.

Under sampled MRI Data



Project Objective

Implement and evaluate algorithms that reconstruct high-quality cardiac MRI images from undersampled k-space data, reducing acquisition time without sacrificing diagnostic image quality.

Reconstruction Methods

- Compressed Sensing
- Parallel Imaging
- Deep Learning-Based Approaches

4x – 8x

Acceleration factors tested

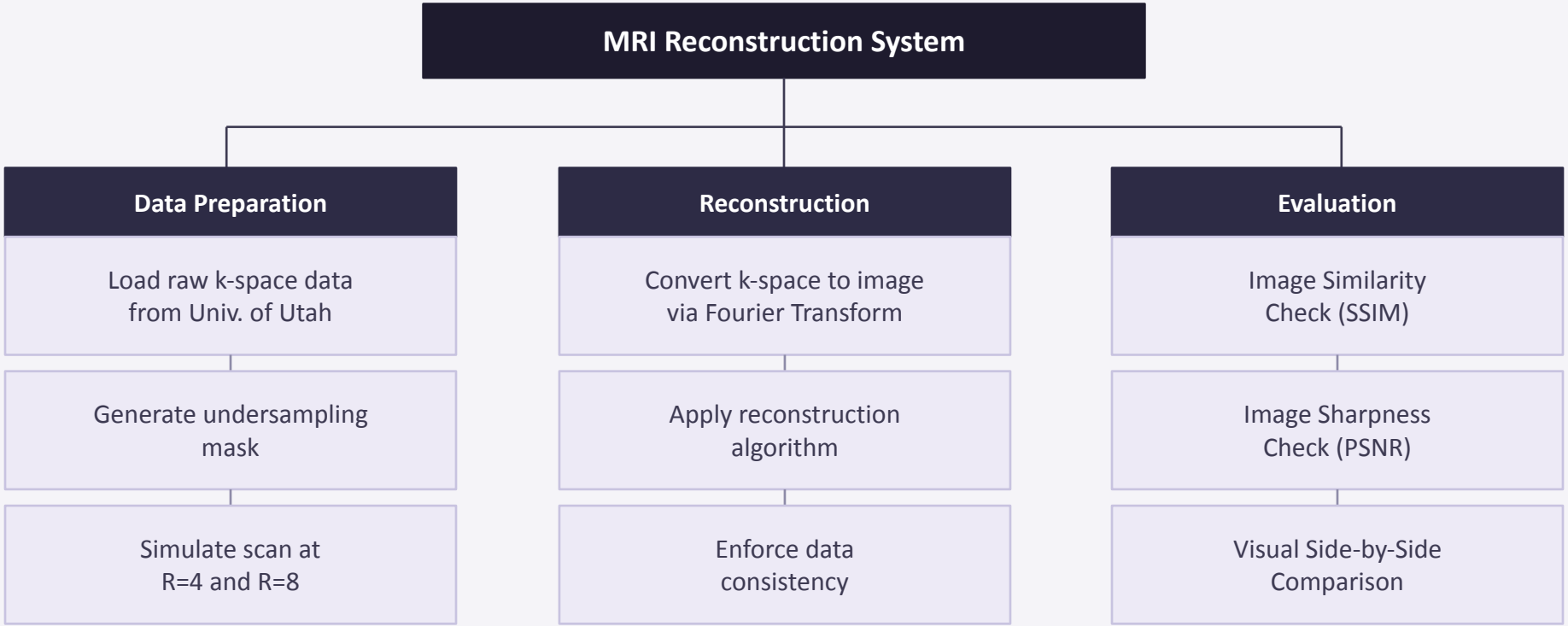
SSIM 0.85

Minimum acceptable score at R=4

2 semesters

~300 hours total project duration

Methodology



System Decomposition

01	**Raw K-Space Data** *Source: University of Utah | Backup: fast MRI (NYU)*	Cardiac MRI k-space data. Sequence type (T1 or T2 weighted) and number of subjects to be confirmed with collaborators.
02	**Undersampling Mask** *Acceleration factor R=4 or R=8 | Pattern: random or center-weighted*	Removes lines from k-space to simulate faster acquisition, with the undersampling rate controlled by the acceleration factor.
03	**Reconstruction Algorithm** *Enforced physics constraints | Toolbox: BART (open-source)*	Rebuilds images from incomplete k-space data. Implementation begins with simple methods and escalates to regularized deep learning.
04	**Image Quality Metric** *SSIM score | PSNR value*	Quantitatively compares reconstruction output to the original fully-sampled MRI image. Target: SSIM >= 0.85 at R=4.
05	**Validation** *Visual comparison | Dice score from segmentation*	Expert radiologist review of reconstructed images for clinical acceptability and absence of visible artifacts.

Commercial Feasibility

Existing Commercial Products

Siemens Deep Resolve

Deep learning reconstruction built into Siemens scanner hardware. Requires purchase of specific scanner models.

GE AIR Recon DL

Integrated into GE imaging systems only. Not available as standalone software.

Both products are proprietary, hardware-dependent, and inaccessible to institutions that cannot afford specific vendor equipment.

This Project

Software-only

Runs on existing hardware. No new equipment required.

Vendor-agnostic

Works with MRI data from any scanner, not tied to one manufacturer.

Open tools

Uses Python, the BART toolbox, and publicly available datasets (fast MRI, NYU).

Low cost

Total project cost under $70, accessible to any researcher or institution.

SIEMENS Healthineers

Strong denoising for fastest acquisitions

Deep Resolve Boost

What is it?

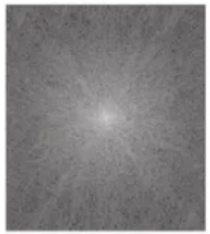
Deep Resolve Boost delivers especially strong denoising for fast acquisitions from head to toe – enabled by Siemens Healthineers' first raw data to image deep learning reconstruction.

How does it work?

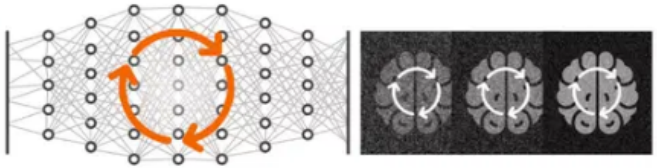
Deep Resolve Boost uses raw data from a reduced, and thus faster scan as input.

In an iterative process, a deep neural network is applied multiple times to generate the final output – with significantly reduced noise. The integration of the raw data along the entire reconstruction process leads to an unmatched performance.

Raw data input from accelerated acquisition



Iterative deep learning reconstruction



Final image output



What is cool about it?



What problem does it solve?



With conventional reconstruction methods, a highly accelerated image acquisition will lead to a strong noise contamination.

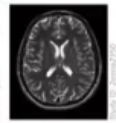


The reconstruction with Deep Resolve Boost enables the generation of images with insanely high SNR and a superfast image acquisition simultaneously.

Clever combination:

Deep Resolve, Boost, Sharp & SMS:
Knee imaging in

88%
shorter scan time: 18 seconds





The benefits:



Unrestricted © Siemens Healthineers, 2022

Testing and Verification

Test 1

Image Quality

SSIM >= 0.85 at R=4

The reconstructed image is compared to the original fully-sampled MRI using SSIM (Structural Similarity Index)

Test 2

Mask Verification

Within 1% of target acceleration factor

Confirms that the undersampling mask removes the correct fraction of k-space lines. The actual acceleration must match the target (R=4 or R=8) to within 1%.

Test 3

Visual Inspection

Expert radiologist review

Side-by-side comparison of the reconstructed and original MRI images, reviewed by a radiologist or experienced MRI researcher for visible artifacts and clinical acceptability.

Summary

This project is feasible, low-cost, and clinically relevant.

- Standard MRI is too slow for cardiac imaging — undersampling is a proven solution
- Reconstruction algorithms can recover image quality from sparse k-space data
- The approach is software-only, vendor-agnostic, and costs less
- Three rigorous verification tests ensure clinical-grade reconstruction quality
- Outcome: a working pipeline that could support future adoption at the University of Utah

Questions?

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